GATE-AII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS



Lecture No.- 02

By- Rahul sir

Recap of Previous Lecture



Topic Fundamental graphs

$\{|x|, \sin x, \cos x, \log x, e^x, x^2, x^3$

Fundamental graphs

Topics to be Covered



Topic Fundamental graphs

Topic Existence of limit

Topic How to evaluate the limit

Greatest Integer of x :

$y = f(x) = [x]$

$f(x) = y = [x]$ → Integer value — defined

$x = 1.1$ $y = [1.1] = 1$

$x = 1.2$ $y = [1.2] = 1$

$x = 1.9$ $y = [1.9] = 1$

$x = 1.99999999$

$y = [1.99999999]$

$= 1$

$x = 0.999999$

$[x] = 0$

$x = 2.9999999$

$[x] = 2$

$f(x) = [x]$ → Piecewise Function

greatest integer Function

Number = Integer + Fractional Part

$5.3 = 5 + .3$

$11.2 = 11 + .2$

$11 = 11 + 0$

Integer value — fractional value

$y = [x]$

x input Integer value → output Also Integer value

$y = [x]$ = greatest Integer Function



2) $f(x) = y = \{x\}$ = Fractional Part of x

Number x = g.I.F + Fractional Part of x

$x = [x] + \{x\}$

$\{x\}$ = Fractional Part of x = $x - [x]$

Defined on Integers

$0 \le x \le 1$ $2 \le x < 3$
$1 \le x < 2$

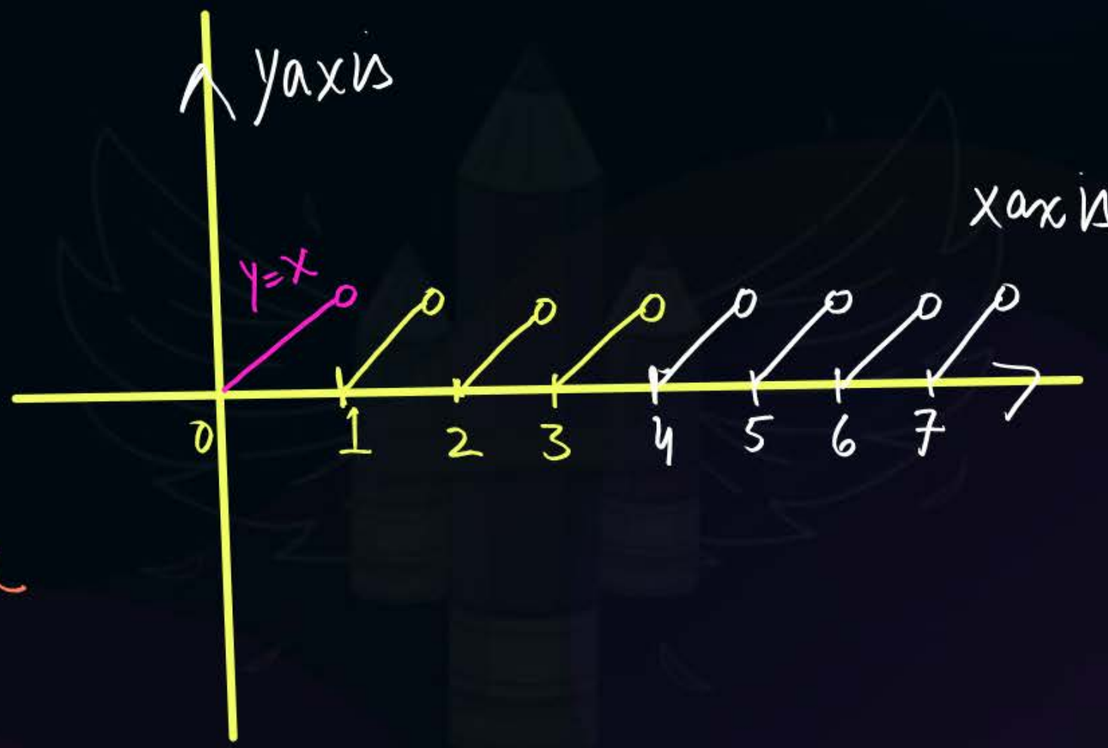
$f(x) = x - [x] = x - 0$ $\quad 0 \le x < 1$

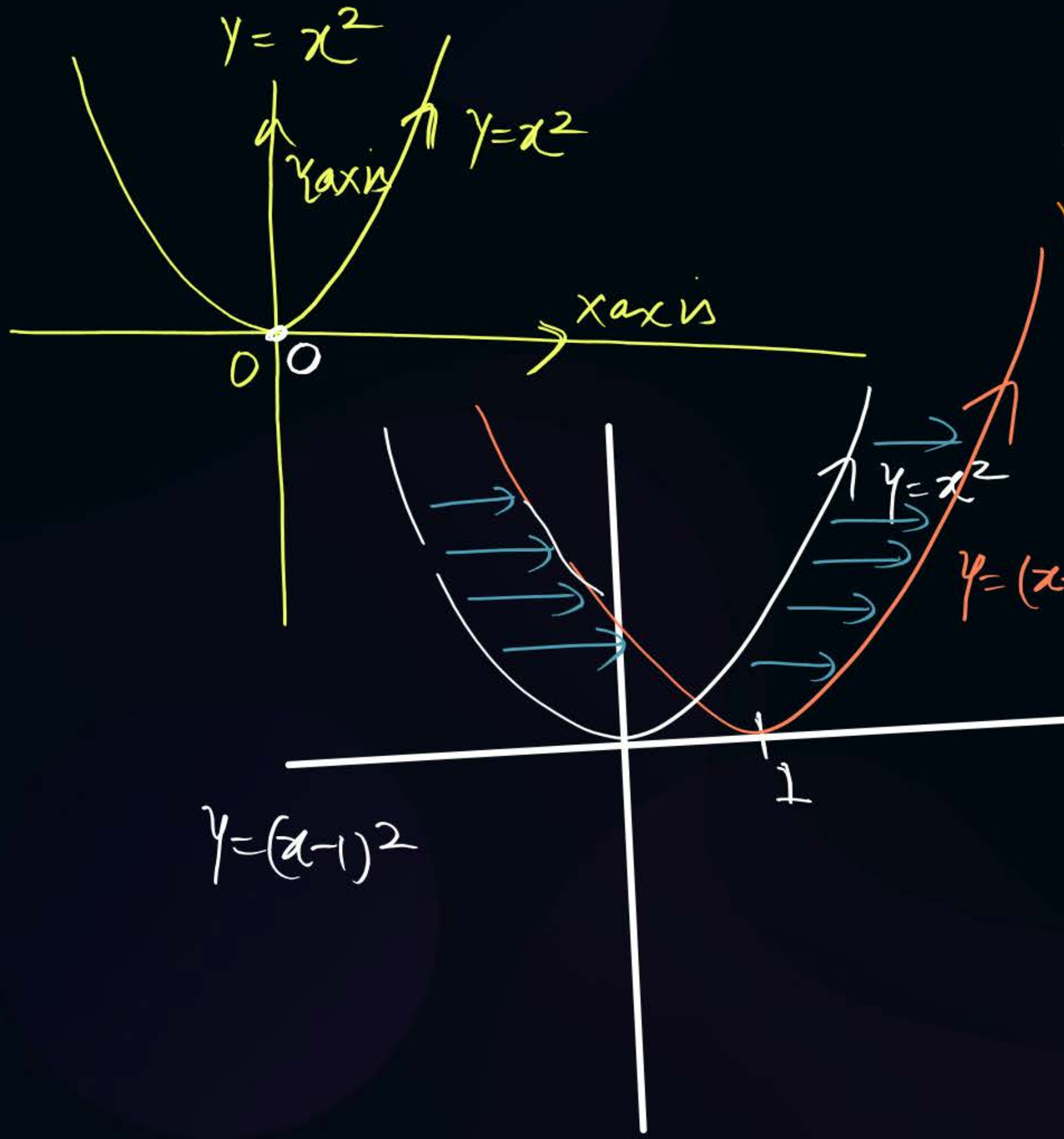
$= x - 1$ $\quad 1 \le x < 2$

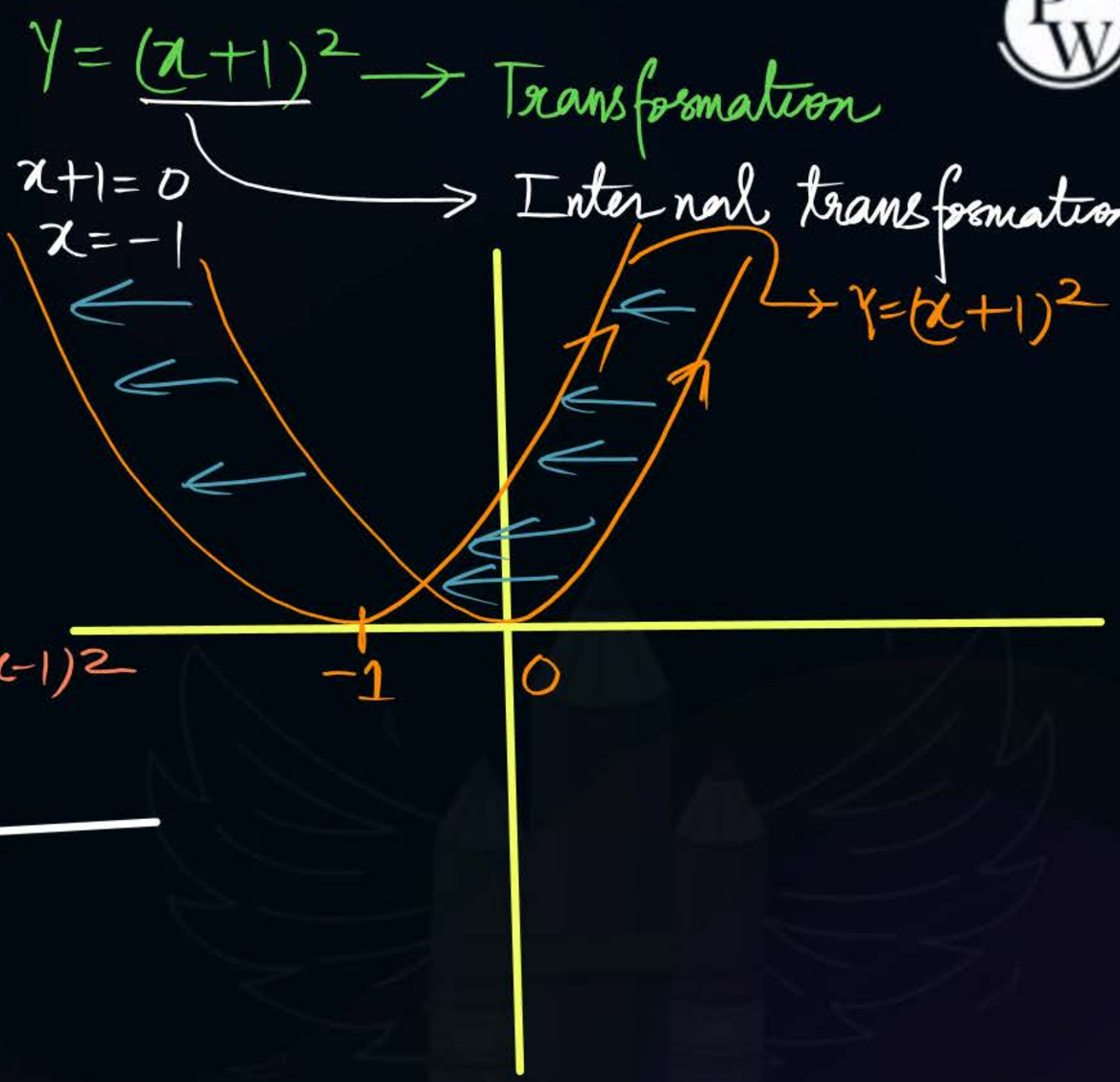
$= x - 2$ $\quad 2 \le x < 3$

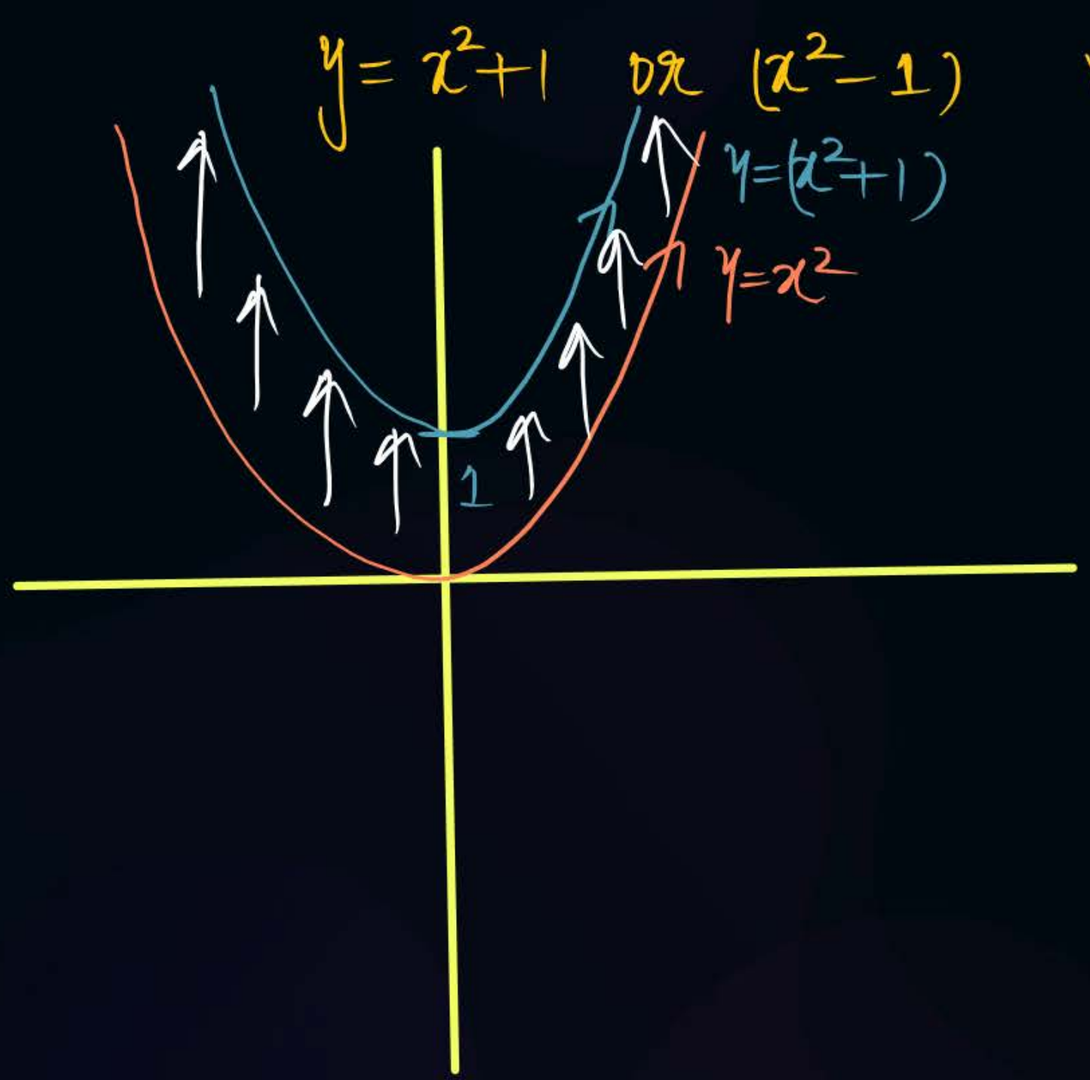
$= x - 3$ $\quad 3 \le x < 4$

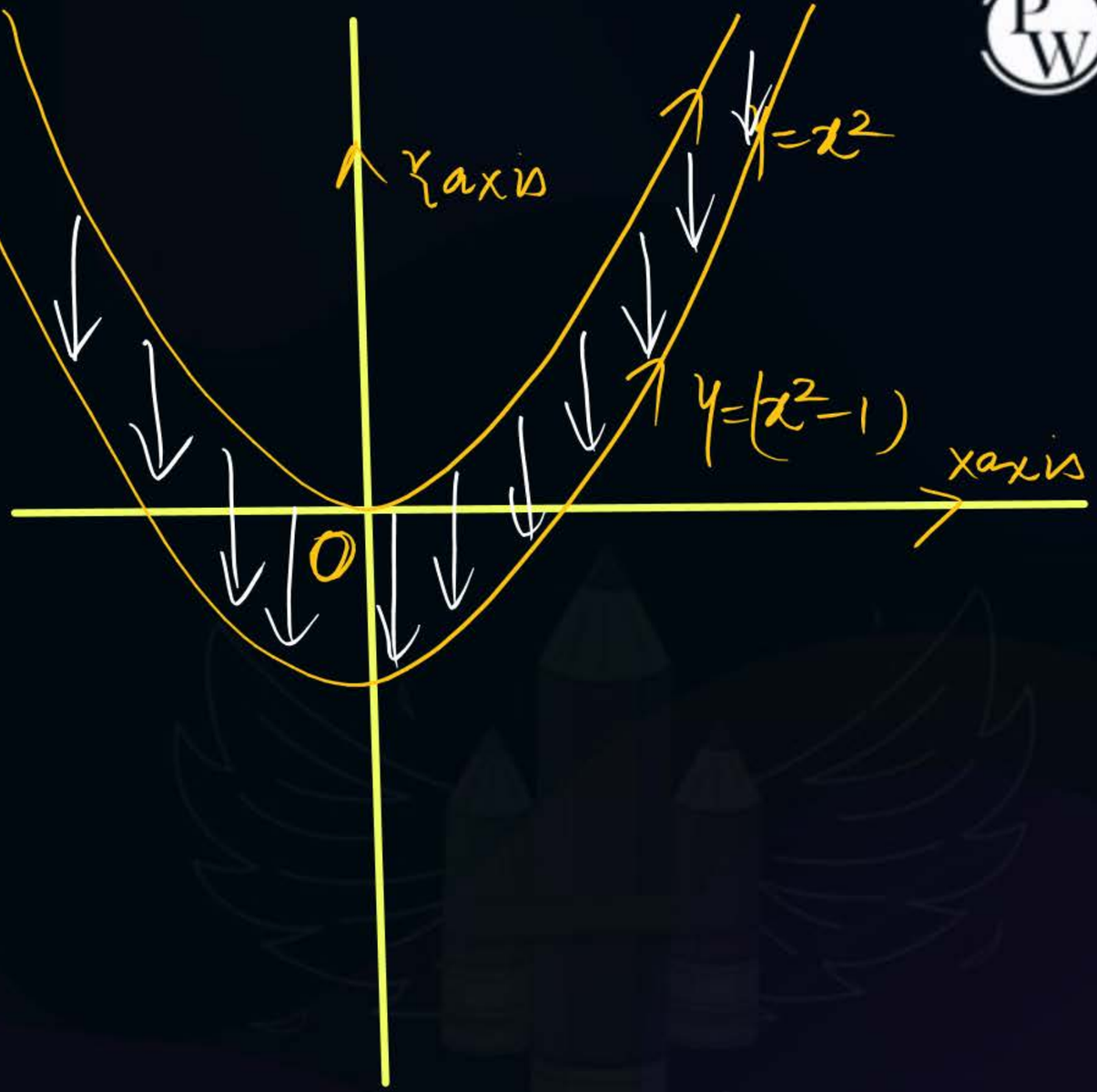
$= x - (n-1)$ $\quad n-1 \le x < n$

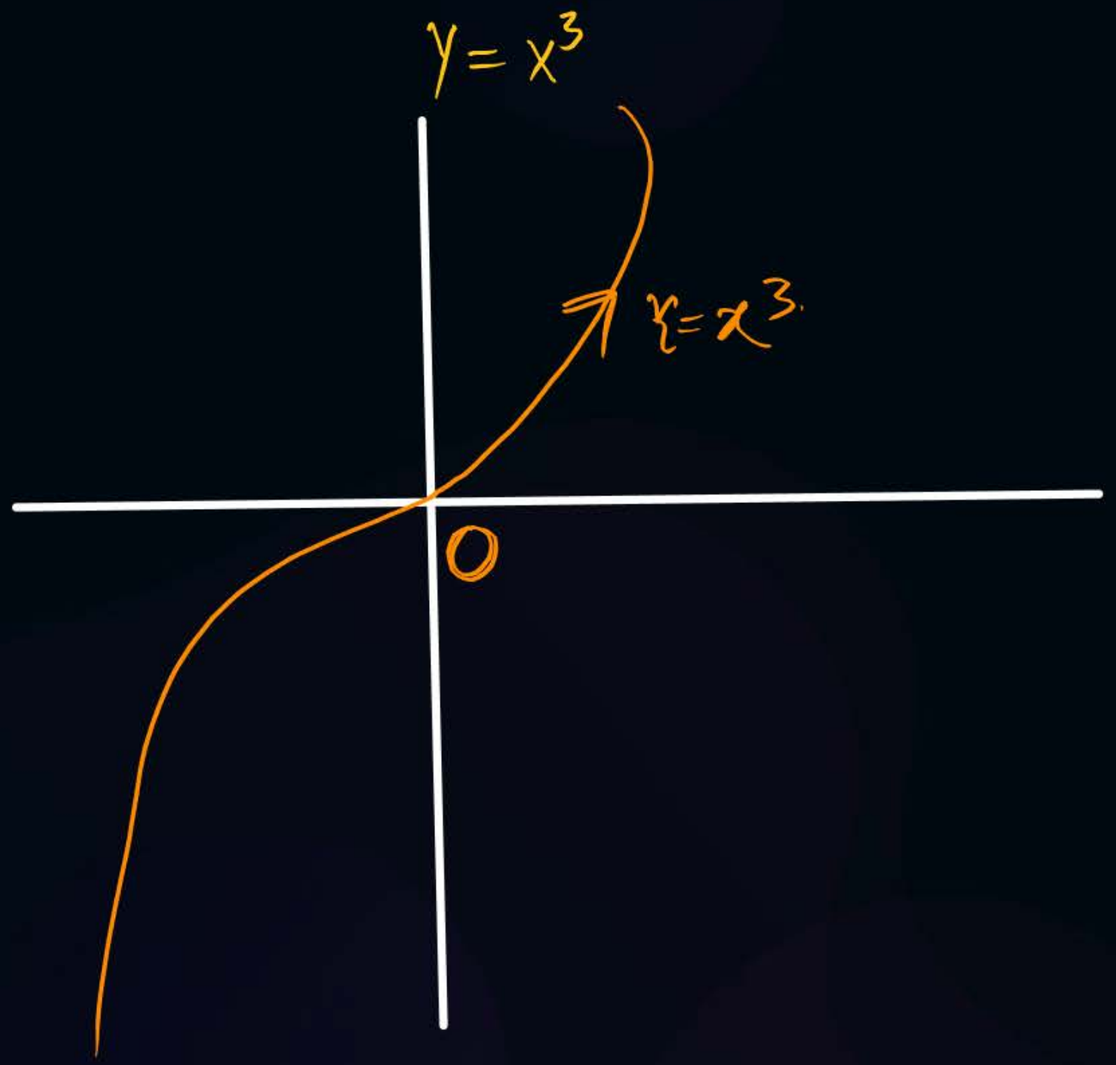



$y = x^2$
yaxis
$y = x^2$
xaxis
0 0
$y = x^2$
$y = (x-1)^2$
1
$y = (x-1)^2$


$y = (x+1)^2$ → Transformation
$x+1 = 0$
$x = -1$
→ Internal transformation
$y = (x+1)^2$
-1
0


$y = x^2+1$ or (x^2-1)
$y=(x^2+1)$
$y=x^2$
1


y axis
$y=x^2$
$y=(x^2-1)$
x axis
O



Cubic Polynomial

Limits and continuity:

$\lim_{x \to a} f(x)$ = Limiting value of Function

x move Towards (a) Finite value.

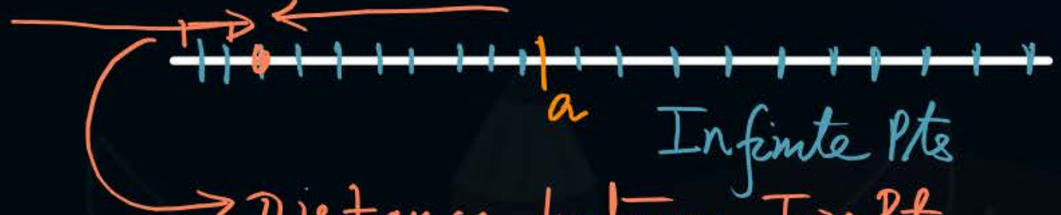
$x \to \to \to a$





Existence of Limit:

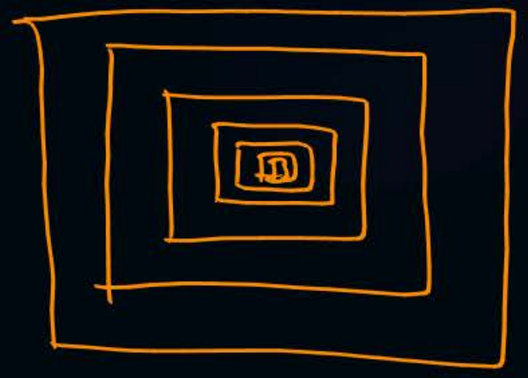




→ Distance between Two Pt

$h > 0$

Step size $h > 0$ → Infinitesimal No.

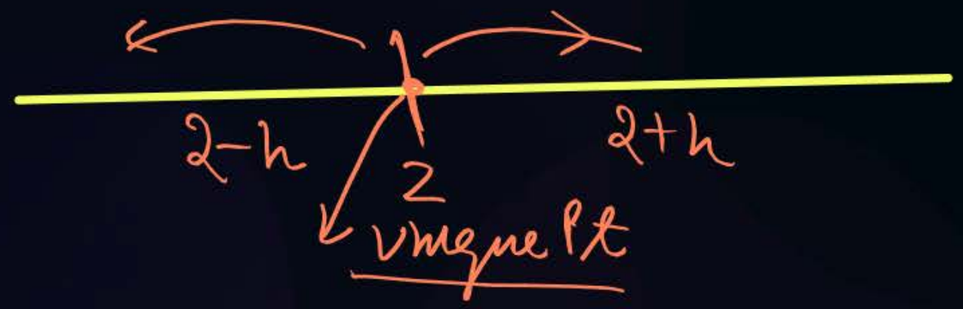


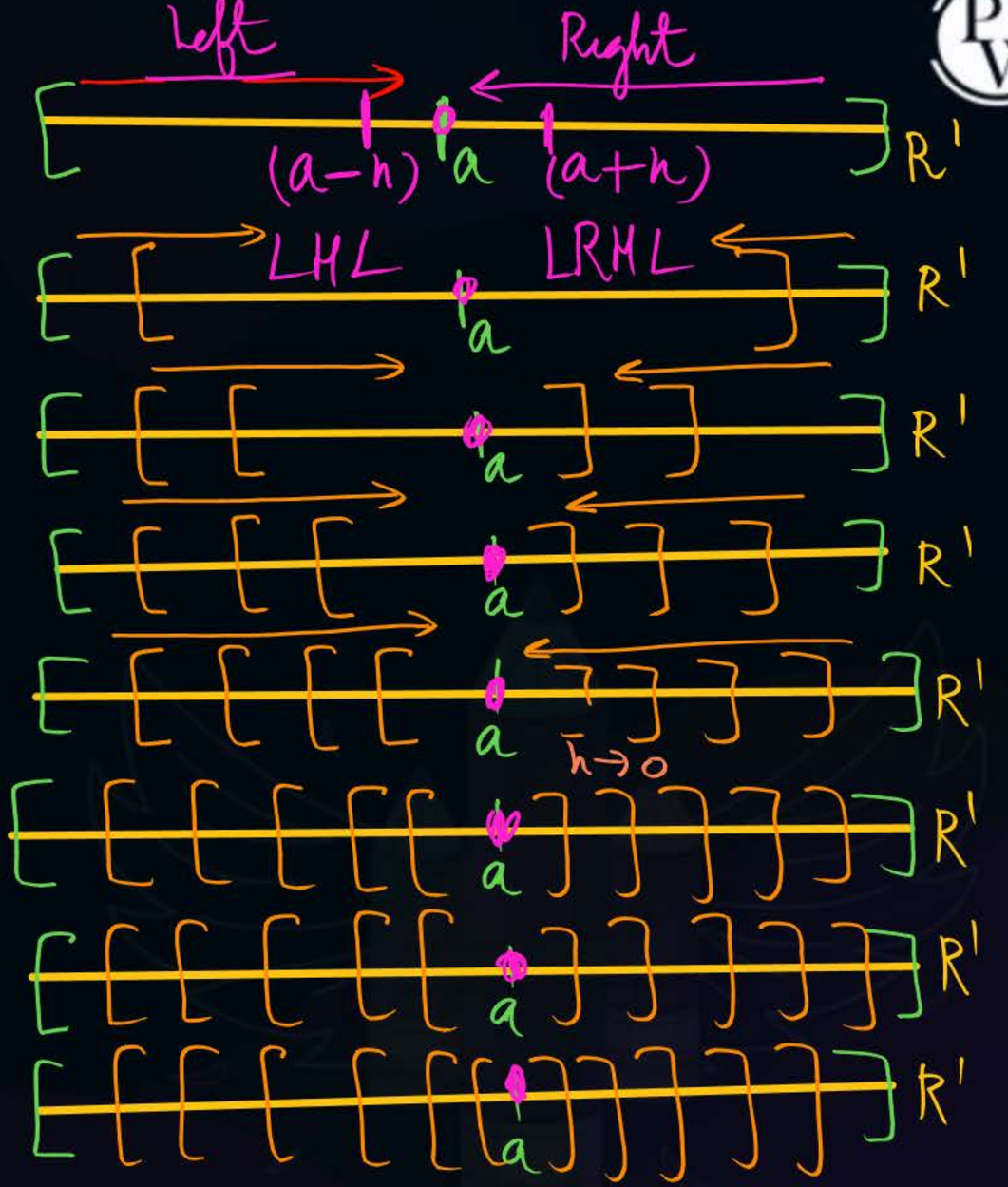
Nested Intervals

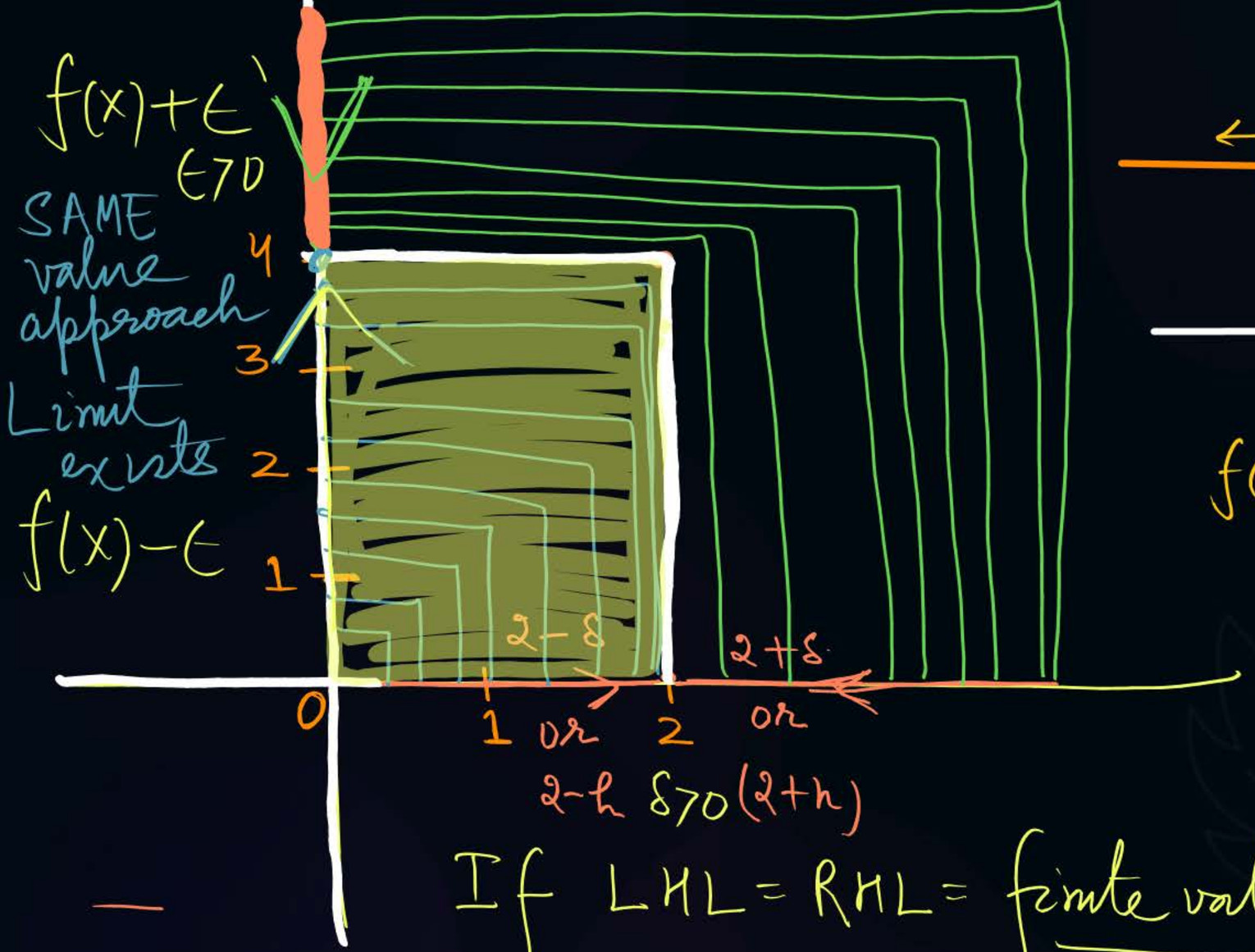


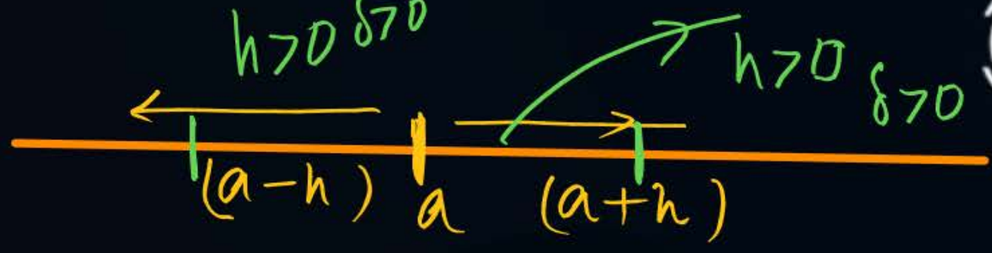
Left Hand Limit $(a-h)$

Right ——— $(a+h)$











$f(x) = y = x+2$

$\lim_{x \to 2} (x+2)$

If LHL = RHL = finite value] Limit exists

If $y=f(x)$ Then Limit exists

LHL = RHL = Fimte value.

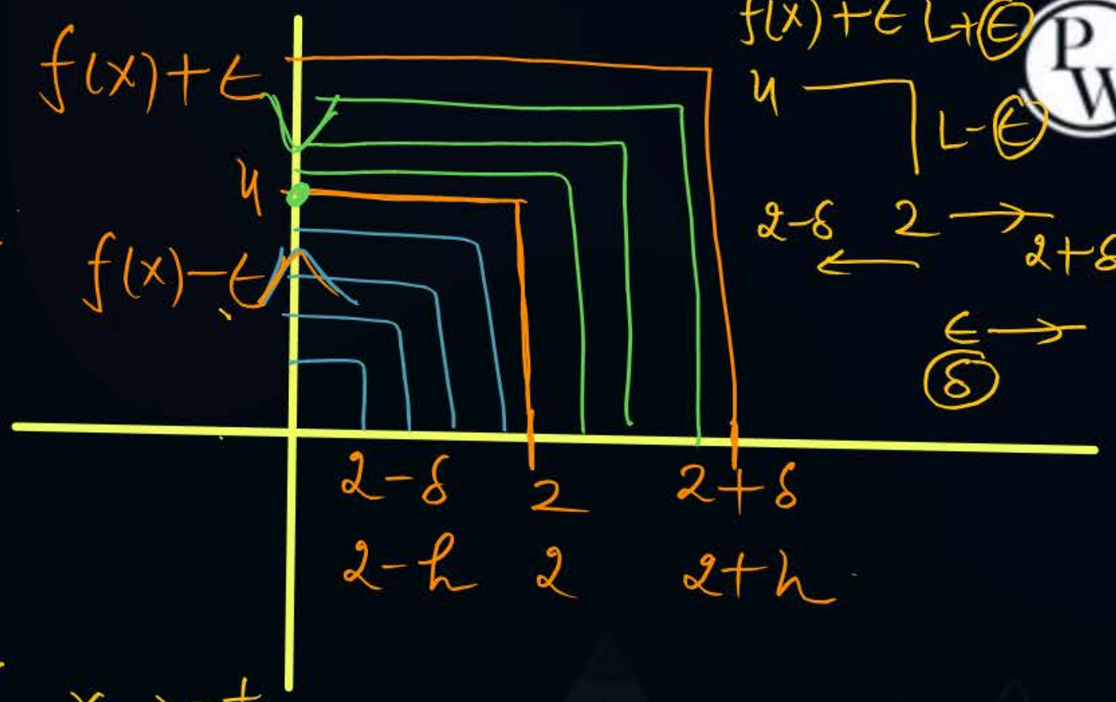
$f(x) = x+2$

Left Hand $= 2^-$

$x \to 2^-$	$f(x)=(x+2)$
1.99	$1.99+2=3.99$
1.999	$1.999+2=3.999$
1.9999	$1.9999+2=3.9999$

Increasing

Big



$x \to 2^+$

$x \to 2^+$	$x+2$
2.01	$2.01+2=4.01$
2.001	$2.001+2=4.001$
2.0001	4.0001

Small

$f(x) = \lim_{x \to 2} (x+2)$

$= 4$

LHL = Left Hand Limit = $\lim_{h\to 0} f(a-h)$ = Finite

RHL = Right Hand Limit = $\lim_{h\to 0} f(a+h)$ = Finite

Existence of Limit # $f(x) = (x+2)$

$\lim_{x\to 2} (x+2)$ $\quad a = 2$

RHL = LHL

Left Hand limit = $\lim_{h\to 0} f(a-h) = \lim_{h\to 0} f(2-h) = \lim_{h\to 0} (4-h) = 4$

$f(x) = (x+2)$

$f(2-h) = (2-h)+2$

$= (4-h)$

Right Hand limit = $\lim_{h\to 0} f(a+h) = \lim_{h\to 0} f(2+h) = \lim_{h\to 0} (4+h)$

$= 4$

LHL = RHL

limit exists



Topic : Single variable calculus

#Q. Show that the limit of : $f(x)=\begin{cases}2x-1 & ; \quad x\le 1 \\ x & ; \quad x>1\end{cases}$ at x = 1 exists.

Left

right

$x \le 1$	$x > 1$
LHL	1 RHL

Left Hand limit

$$= \lim_{h\to 0} f(a-h)$$

$$= \lim_{h\to 0} f(1-h) \quad (2x-1)$$

$$= \lim_{h\to 0} [2(1-h)-1]$$

$$= \lim_{h\to 0} [2-2h-1]$$

$$= 1$$

$$RHL = \lim_{h\to 0} f(a+h)$$

$$= \lim_{h\to 0} f(1+h) \quad x \to (1+h)$$

$$= \lim_{h\to 0} (1+h)$$

$$= 1$$

LHL = RHL

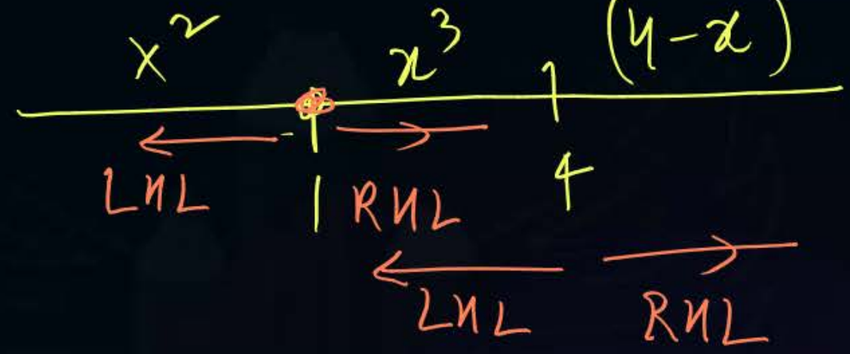


Topic : Single variable calculus

1 ✓
4 X

#Q. Evaluate the left hand and right-hand limits of the function

defined by $f(x)=\begin{cases} x^2 & , \quad x<1 \\ x^3 & , \quad 1<x<4 \\ 4-x & , \quad x>4 \end{cases}$ at x = 1, 4 and hence



check existence of limit at x = 1, 4.

$x=1$ exists
$x=4$ does Not exists]

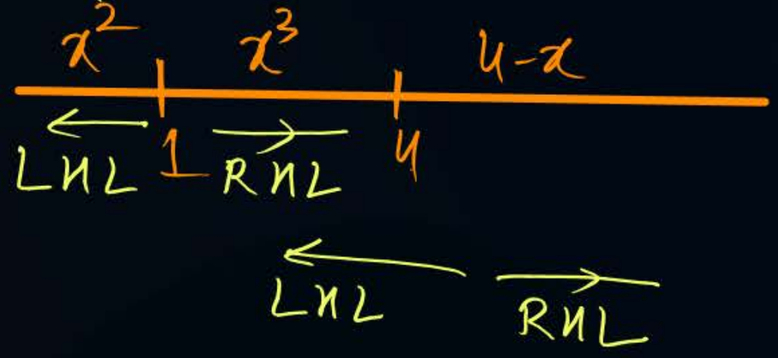
Limit existence $x=1$

$$LHL = \lim_{h\to 0} f(1-h) = \lim_{h\to 0} (1-h)^2$$

$$= \lim_{h\to 0} (1+h^2+2h) = 1$$

$$RHL = \lim_{h\to 0} f(1+h) = \lim_{h\to 0} (1+h)^3 = \lim_{h\to 0} (1+h^3+3h^2+3h] = 1$$

$\boxed{LHL = RHL \text{ at } x=1}$ exists The Limit ✓



Limit existence $x=4$

$$LHL = \lim_{h\to 0} f(4-h) = \lim_{h\to 0} (4-h)^3 = 64$$

$$RHL = \lim_{h\to 0} f(a+h) = \lim_{h\to 0} f(4+h) = \lim_{h\to 0} (4-(4+h)] = 0$$

$LHL \neq RHL$ Does Not exists



Topic : Single variable calculus

does Not exists

#Q. Evaluate the left hand and right-hand limits of the function

$$f(x)=\begin{cases}\dfrac{\sqrt{(x^2-6x+9)}}{(x-3)}, & x\neq 3\\ 0, & x=3\end{cases}$$

$\sqrt{(x^2-6x+9)} \rightarrow \sqrt{(x-3)^2}$



at x = 3 and hence comment on the existence of limit at x = 3.

$$f(x)=\begin{cases}\dfrac{\sqrt{(x-3)^2}}{x-3}, & x\neq 3\\ 0, & x=3\end{cases}$$

$\sqrt{x^2}=|x|$

$|-5| = 5$

$$f(x) = \begin{cases} \frac{\sqrt{(x-3)^2}}{(x-3)} & x \neq 3 \\ 0 & x = 3 \end{cases} = \begin{cases} \frac{|x-3|}{(x-3)} & x \neq 3 \\ 0 & x = 3 \end{cases}$$

does Not exists

Left Hand Limit $= \lim_{h \to 0} f(a-h) = \lim_{h \to 0} f(3-h)$

$$= \lim_{h \to 0} \frac{|3-h-3|}{[(3-h)-3)} = \lim_{h \to 0} \frac{|-h|}{-h} = \lim_{h \to 0} \frac{h}{-h} = -1$$

Right Hand limit $= \lim_{h \to 0} f(a+h) = \lim_{h \to 0} f(3+h)$

$$= \lim_{h \to 0} \frac{|3+h-3|}{(3+h-3)} = \lim_{h \to 0} \frac{|h|}{h} = 1$$

Plug-in

LHL ≠ RHL

✓ (romit)

LHL ≠ RHL

✓ Square Root Function ✓

✓ Modulas Function

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always does Not exists

THANK - YOU